DBMS & SQL – Notes

What is DBMS?

DBMS is software used to manage data in an organized way.

Functions discussed:

- Store data
- Update data
- Delete data
- Secure data
- Retrieve data

Examples:
MySQL, Oracle, SQL Server, MongoDB

Types of DBMS

1. **Hierarchical DBMS**
   - Tree structure
   - Example: Company → Employees
2. **Network DBMS**
   - Many-to-many relationships
   - Example: Students ↔ Courses
3. **Relational DBMS (RDBMS)**
   - Data in tables
   - Example: Employee database
4. **NoSQL DBMS**
   - Non-table format
   - Example: JSON data storage

SQL vs MySQL

- SQL = Query language
- MySQL = Database software

Simply:
SQL = Language
MySQL = Tool

Basic Concepts

Database

Collection of data
Example: Library database

Table

Stores data in rows & columns

Example: Books Table

book_id	title	author
1	Java Basics	Arun
2	DBMS Guide	Neha

Columns (Fields)

- book_id
- title
- author

Rows (Records)

Each row = one book

Values

Actual data like "Java Basics"

Data Types

- Numeric → INT, DECIMAL
- String → VARCHAR, TEXT
- Date → DATE
- Boolean → TRUE/FALSE

Constraints

- NOT NULL → must have value
- UNIQUE → no duplicates
- PRIMARY KEY → unique identifier
- FOREIGN KEY → relation between tables
- CHECK → condition
- DEFAULT → automatic value

SQL Commands

1. CREATE

```
CREATE DATABASE library;
CREATE TABLE books (
  book_id INT PRIMARY KEY,
  title VARCHAR(100),
  author VARCHAR(50),
  price DECIMAL(5,2)
);
```

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2. USE

```
USE library;
```

---

3. INSERT

```
INSERT INTO books VALUES (1, 'Java Basics', 'Arun', 299.99);
INSERT INTO books VALUES (2, 'SQL Made Easy', 'Neha', 199.50);
```

---

4. SELECT

```
SELECT * FROM books;
SELECT title, price FROM books;
```

---

5. FROM

```
SELECT * FROM books;
```

Example

```
CREATE DATABASE library;

USE library;
```

```sql
CREATE TABLE books (
  book_id INT,
  title VARCHAR(100),
  author VARCHAR(50),
  price DECIMAL(5,2)
);

INSERT INTO books VALUES (1, 'Java Basics', 'Arun', 299.99);
INSERT INTO books VALUES (2, 'SQL Made Easy', 'Neha', 199.50);

SELECT * FROM books;
```

Output

book_id	title	author	price
1	Java Basics	Arun	299.99
2	SQL Made Easy	Neha	199.50